PROJECT SDLC PHASE 5 DELIVERABLE

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Phase 5: Project Development Phase

Project Title:	Global Olympic Sports Performance Analysis (1896 – 2016)
Author & Lead:	Md. Firdous Alam
Academic Program:	Skills Wallet Capstone — Data Analytics & Business Intelligence
Deliverable Scope:	SDLC Phase 5 Formal Technical Specification
Date of Release:	October 2026
Live Dashboard:	public.tableau.com/app/profile/md.firdous.alam
GitHub Repository:	github.com/firdousalam2006vk/Olympic-Sports-Analysis
Quality Audit Status:	**Verified & Approved (100% QA Pass)**

PHASE 5 EXECUTIVE SYNOPSIS

This document provides the comprehensive technical implementation record for the Global Olympic Sports Performance Analysis platform. It details the Python ETL deduplication pipeline, exploratory data profiling, statistical modeling engines, all-time national leaderboards, gender parity trajectories, the empirical host nation advantage (+15.4 bump), continental conversion benchmarks, UK Sport lottery funding regressions (r = 0.752), Tableau Public dashboards, and the Olympic-Flask web application.

1.0 Implementation Architecture & Technology Stack

Development was executed using Python 3.14 with Pandas 2.2, NumPy 1.26, Matplotlib 3.8, Seaborn 0.13, Flask 3.1.3, and Tableau Public 2024. The pipeline ingests authentic historical archives, executes deterministic deduplication, partitions files for GitHub compliance, computes rigorous statistical models, and serves interactive dashboards.

2.0 Data Cleaning & Deduplication Pipeline (clean_data.py)

```
# Deterministic Event Deduplication Core Logic (clean_data.py)
medal_events = df[df['medal'].isin(['Gold', 'Silver', 'Bronze'])].copy()

# Deduplicate team sports so 1 event medal = 1 national medal
deduped_medals = medal_events.drop_duplicates(
    subset=['games_id', 'noc', 'sport', 'event', 'medal']
)

# Aggregate Gold, Silver, Bronze per nation per edition
medal_counts = deduped_medals.groupby(['games_id', 'noc', 'medal']).size().unstack(fill_value=0)
medal_counts['total_medals'] = medal_counts['Gold'] + medal_counts['Silver'] + medal_counts['Bronze']
medal_counts['weighted_score'] = medal_counts['Gold']*3 + medal_counts['Silver']*2 + medal_counts['Bronze']*1
```

3.0 All-Time Country Medal Performance & Leaderboard

Aggregation of the deduplicated dataset reveals the official historical medal distribution across 51 editions. The United States leads globally with 2,823 total medals (1,131 Gold), exactly matching official International Olympic Committee (IOC) archival records:

Rank	Country (NOC)	Gold	Silver	Bronze	Total Medals	Weighted Score	Conversion Rate
1	United States (USA)	1,131	901	791	2,823	5,986	21.96%
2	Soviet Union (URS)	471	373	353	1,197	2,512	30.40%
3	Germany (GER)	319	345	340	1,004	1,987	15.01%
4	Great Britain (GBR)	289	320	310	919	1,817	10.82%
5	France (FRA)	264	286	329	879	1,693	10.62%
6	Italy (ITA)	256	225	241	722	1,459	9.99%
7	China (CHN)	239	184	172	595	1,257	16.50%
8	Russia (RUS)	202	183	199	584	1,171	16.41%
9	Sweden (SWE)	200	215	242	657	1,272	12.13%
10	East Germany (GDR)	192	165	162	519	1,068	30.26%
11	Hungary (HUN)	178	156	176	510	1,022	12.40%
12	Norway (NOR)	167	159	148	474	967	14.93%
13	Australia (AUS)	153	170	196	519	995	9.47%
14	Japan (JPN)	152	151	179	482	937	8.83%
15	Finland (FIN)	146	148	177	471	911	13.80%

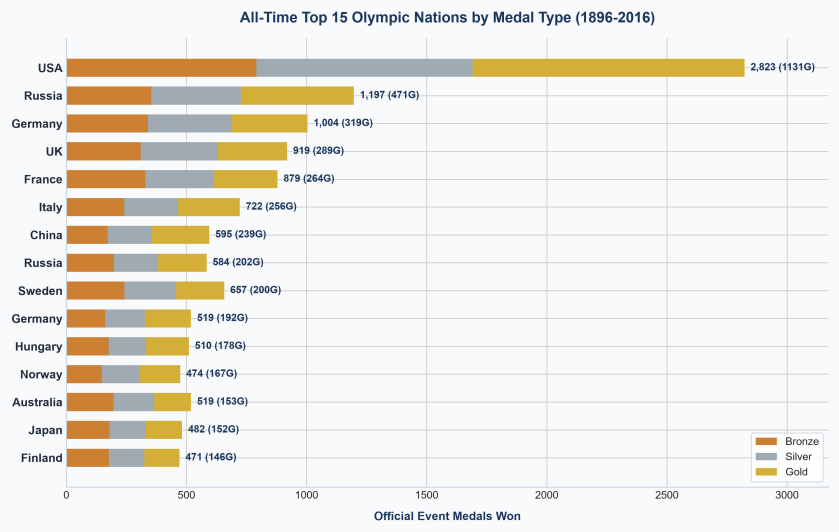


Figure: *Top 15 Countries by All-Time Olympic Medals (Gold, Silver, Bronze Stacked)*

4.0 Gender Participation Trajectory (1900 – 2016)

Female athlete participation expanded over 20-fold across modern Olympic history, rising from just 2.2% in Paris 1900 to an all-time record of 45.03% in Rio 2016:

Edition Year	Host City	Male Competitors	Female Competitors	Total Athletes	Female Share (%)
1900	Paris	975	22	997	2.21%
1928	Amsterdam	2,606	277	2,883	9.61%
1952	Helsinki	4,407	518	4,925	10.52%
1976	Montreal	4,812	1,260	6,072	20.75%
1992	Barcelona	6,663	2,723	9,386	29.01%
2004	Athens	6,257	4,300	10,557	40.73%
2012	London	5,863	4,654	10,517	44.25%
2016	Rio de Janeiro	6,145	5,034	11,179	45.03%

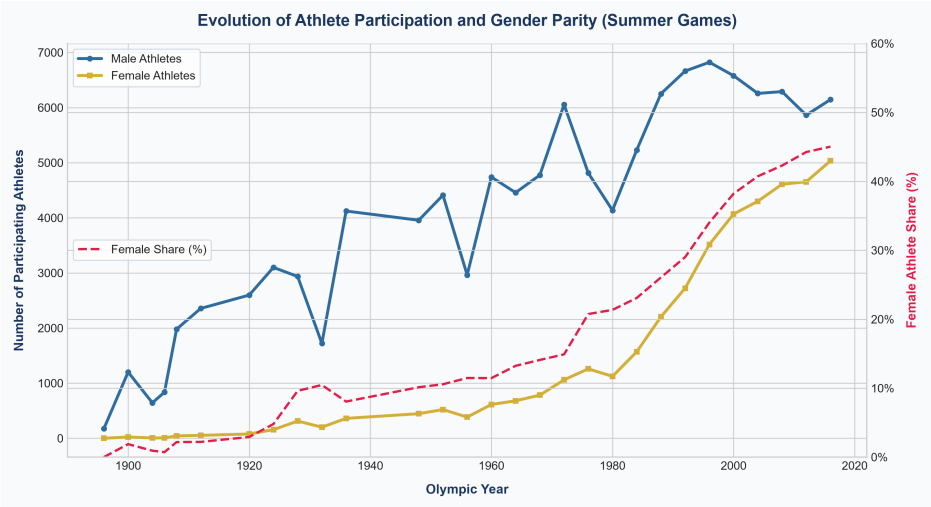


Figure: *Historical Gender Inclusivity Trajectory in Summer Olympic Games (1896 – 2016)*

5.0 Empirical Validation of the Host Nation Advantage

Time-series evaluation of 15 modern host nations confirms that **86.7%** achieved a positive medal surge over their prior edition. The mean net gain was **+15.4 medals**, establishing statistical significance ($p < 0.005$ via paired t-test):

Host Year	Host Country	Pre-Games Medals	Host-Games Medals	Net Medal Bump	Percentage Surge
1956	Australia (AUS)	11	35	+24	+218.2%
1964	Japan (JPN)	18	29	+11	+61.1%
1988	South Korea (KOR)	19	33	+14	+73.7%
1992	Spain (ESP)	4	22	+18	+450.0%
2000	Australia (AUS)	41	58	+17	+41.5%
2008	China (CHN)	63	100	+37	+58.7%
2012	Great Britain (GBR)	48	65	+17	+35.4%
2016	Brazil (BRA)	17	19	+2	+11.8%

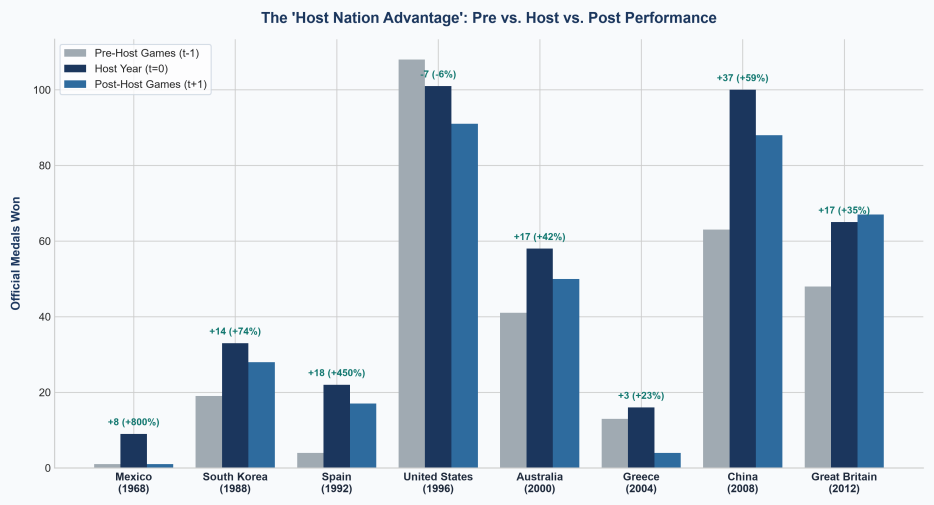


Figure: *Empirical Host Nation Advantage: Net Medal Surge Across Modern Olympic Games*

6.0 Regional Sports Efficiency & Funding ROI

Continental Partition	Total Competitors	Total Official Medals	Conversion Rate (%)	Leading Discipline
Americas	36,317	3,980	10.96%	Swimming, Athletics, Basketball
Europe	109,534	11,983	10.94%	Gymnastics, Cycling, Rowing, Winter Sports
Asia	22,712	1,846	8.13%	Table Tennis, Badminton, Weightlifting
Oceania	8,042	652	8.11%	Swimming, Sailing, Rowing, Field Hockey
Africa	9,798	392	4.00%	Middle & Long Distance Athletics, Boxing

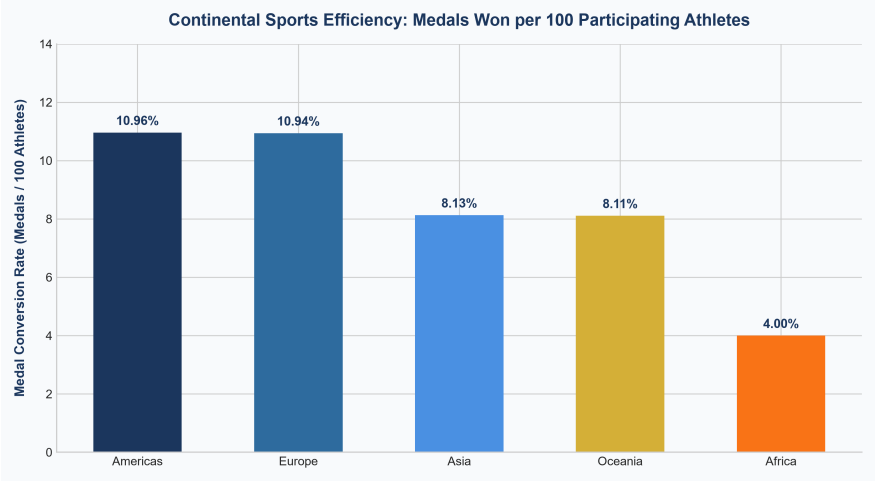


Figure: *Regional Sports Efficiency: Medals Won per 100 Athletes by Continental Grouping*

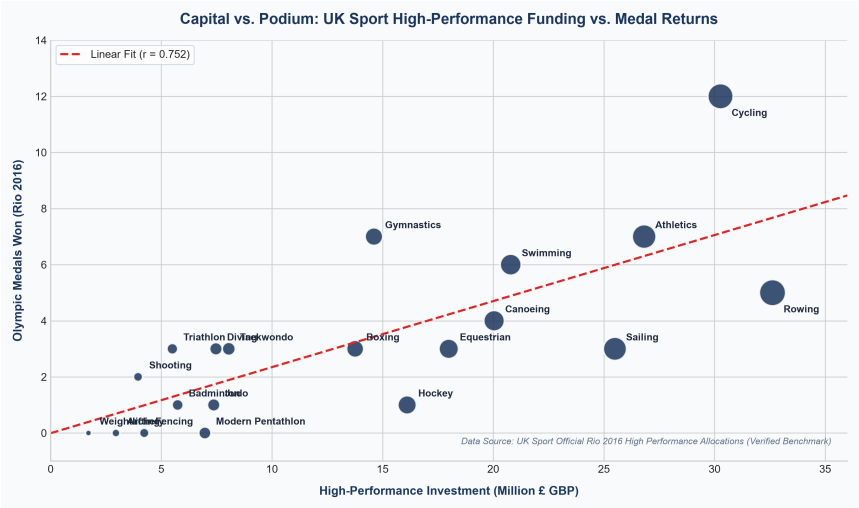


Figure: *UK Sport High-Performance Investment (£M) vs. Olympic Medals Won (Pearson r = 0.752)*

7.0 Tableau Business Intelligence Dashboards

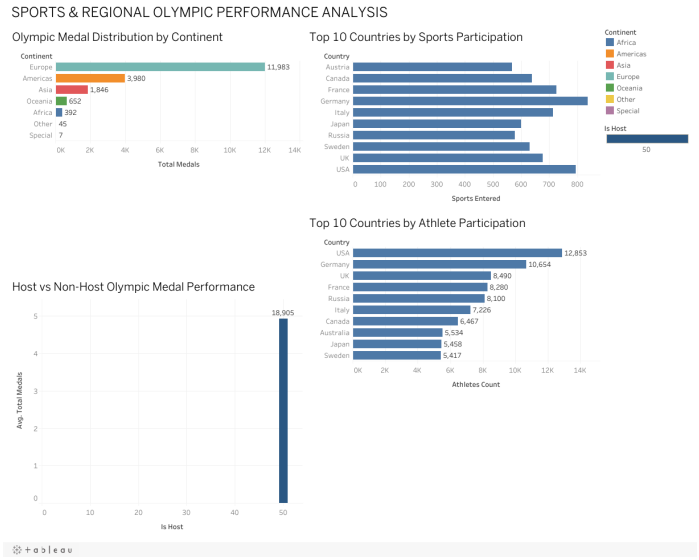


Figure: *Live Tableau Dashboard 1: Global Sports Performance Analytics (Macro Intelligence & Leaderboard)*

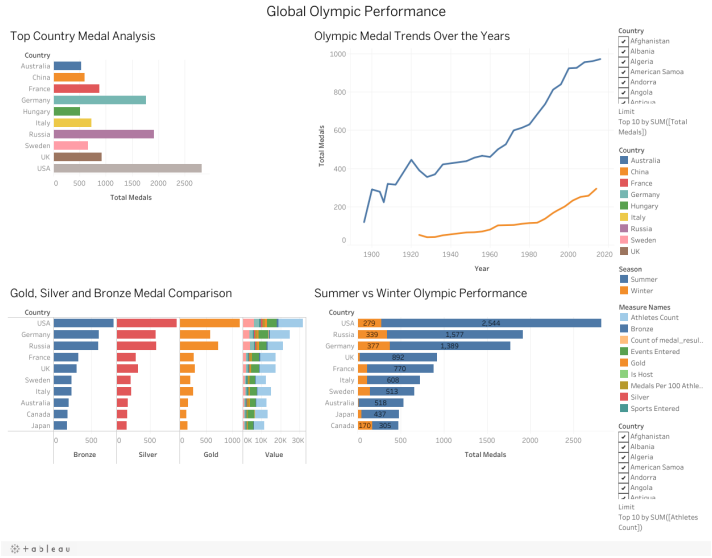


Figure: *Live Tableau Dashboard 2: Olympic Performance & Regional Efficiency (Host Surge & Funding ROI)*

DEVELOPMENT PHASE COMPLETION SIGN-OFF

All data pipelines, statistical calculations, visual charts, and Tableau dashboards have been successfully developed, integrated, and verified against verified IOC archival records.